

Adem Mrabet

Artificial Intelligence Engineer

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PROFESSIONAL SUMMARY

- Master's student in Data Science (Polytech Intl, 2024–2026) who designed and shipped a deployed, full-stack Retrieval-Augmented Generation platform for the African Development Bank's internal governance documentation.
- Built a deterministic extraction pipeline turning an unstructured 250+ page PDF into a 327-node knowledge graph (1,776 responsibility records, 131 distinct roles), with a grounded-generation layer that verifies every LLM-rephrased answer against retrieved facts before it reaches the user.
- Delivered applied ML projects spanning computer vision (92% accuracy CNN), Big Data engineering (73% precision PySpark fraud detection), and predictive analytics (86% accuracy), plus a second RAG system for legal document Q&A.

WORK EXPERIENCE

Artificial Intelligence Engineer — Final-Year Project (PFE), African Development Bank / Polytech Intl

Apr. 2026 – Sept. 2026 | Tunis, Tunisia

- Designed and built an end-to-end knowledge-graph RAG system answering governance questions over the Bank's Delegation of Authority Matrix, starting with no pre-existing dataset or schema.
- Built a deterministic, 8-stage extraction pipeline (character-level PDF parsing with pdfplumber) resolving the source document into a validated graph of 327 activity nodes, 1,776 responsibility edges, and 131 distinct roles, at a 0.6% unresolved-role rate.
- Implemented hybrid retrieval (identifier lookup plus TF-IDF lexical search with deterministic typo correction) and a grounded-generation layer: an LLM only rephrases pre-assembled, graph-sourced answers, and a verification step discards any rephrasing that drops a fact.
- Shipped as a production platform — FastAPI, React, PostgreSQL, JWT + Google/Microsoft OAuth, Docker — with real-time conversation sharing (Server-Sent Events), voice input, attachment Q&A, and an analytics dashboard; deployed and publicly reachable.

IT HelpDesk, Outsourcia (on assignment at the African Development Bank)

Since Sept. 2023 | Tunis, Tunisia

- Resolved 1,000+ IT incidents while maintaining business continuity; configured VPN and Remote Desktop access solutions.
- Built PowerBI dashboards and automated Excel reporting for system monitoring, cutting recurring incidents by ~20% and operational risk by 17%.

Cybersecurity Data Analyst Intern, BH Bank

Feb. 2022 – May 2022 | Tunis, Tunisia

- Built an ETL pipeline with Talend to process and transform large transactional datasets.
- Conducted firewall-log traffic analysis (peak hours, threats, common destinations) and developed PowerBI dashboards for system monitoring and reporting.

EDUCATION

Master's Degree in Data Science, Polytech Intl

Since Oct. 2024 | Tunis, Tunisia

Machine Learning | Deep Learning (CNN, Transfer Learning) | NLP | Statistics | Optimization | Big Data

Bachelor's Degree in Business Intelligence, Jendouba University

Oct. 2019 – Jun. 2022 | Jendouba, Tunisia

Data Warehousing | Programming (Python, Java, C, C#) | Databases (SQL, NoSQL, PL/SQL) | Data Analysis & Visualization

High School Diploma, Farhat Hached High School

Sept. 2015 – Jul. 2019 | Radis, Tunisia

ACADEMIC PROJECTS

Legal Chatbot — Retrieval-Augmented Generation (RAG) System

Since Dec. 2025 | Polytech Intl

Python, HuggingFace, FAISS, BERT, Groq

- Built a full RAG pipeline to answer questions from multi-PDF legal documents: text extraction, cleaning, chunking, and metadata structuring, followed by BERT embeddings indexed in FAISS for semantic search.
- Integrated HuggingFace Transformers with a custom retriever and Groq-based generation to produce structured answers referencing relevant legal articles, in a modular, multi-document-ready architecture.

Skin Lesion Classification

Nov. 2025 | Polytech Intl

TensorFlow, Keras, CNNs, Transfer Learning

- Built a CNN image classifier (VGG16 / ResNet transfer learning) on the HAM10000 dermoscopic dataset, with preprocessing, augmentation, and hyperparameter tuning, achieving 92% accuracy.

Formula 1 Championship Predictor

Mar. 2024 | Polytech Intl

Python, Scikit-Learn, Predictive Analytics

- Built a predictive analytics pipeline forecasting the F1 World Champion from multi-season driver, race, and constructor statistics, achieving 86% accuracy.

Stock Market Forecasting

Nov. 2025 | Polytech Intl

Python, Scikit-Learn, XGBoost

- Built an end-to-end ML pipeline for hourly stock-direction forecasting, with leakage-safe temporal splitting (TimeSeriesSplit), engineered lag/volatility/volume features, and hyperparameter-tuned models (Logistic Regression, Random Forest, XGBoost), reaching a 58% backtested hit rate.

Credit Card Fraud Detection

Oct. 2025 | Polytech Intl

PySpark, Spark MLlib, Big Data Engineering

- Built a distributed fraud-detection pipeline on PySpark/Spark MLlib (ETL, feature engineering, model training and evaluation across Logistic Regression, Random Forest, and Gradient Boosting), reaching 73% precision on large-scale transactional data.

CERTIFICATES

Aspire Leadership Program, Aspire Institute (Harvard-affiliated)

Sept. – Dec. 2025 | Cambridge, MA, USA

- Selected finalist among 9,988 participants worldwide (out of 54,337 applicants); completed 40 hours of Harvard-faculty-led coursework on leadership, critical thinking, and AI's societal impact.

FullStack JS, GoMyCode

Oct. 2022 – May 2023 | Tunis, Tunisia

Fundamentals of Deep Learning, Nvidia

Oct. 2025 | Tunis, Tunisia

SCRUM Master Certification Specialization, LearnQuest

Apr. – May 2025 | Tunis, Tunisia

SKILLS

- **LLMs & GenAI:** RAG, Knowledge Graphs, Prompt/Context Engineering, Groq, HuggingFace Transformers, FAISS & Vector Search, BERT
- **Web & Backend:** FastAPI, React, REST APIs, PostgreSQL, JWT / OAuth2 (Google, Microsoft), Docker, Server-Sent Events
- **Deep Learning:** CNNs, Transfer Learning, TensorFlow, PyTorch, Keras
- **Big Data & Distributed Systems:** PySpark, Spark MLlib, Hadoop, Kafka
- **Cloud & MLOps:** AWS (S3, Lambda, Crawler, Athena), Docker, MLflow
- **Programming & Data:** Python (NumPy, Pandas, Scikit-learn), SQL, R
- **Machine Learning:** Supervised Learning (Classification, Regression), Feature Engineering, Model Evaluation
- **Other Tools:** Git/GitHub, Jupyter, Excel, Microsoft 365
- **Professional:** Communication & Collaboration, Analytical Thinking, Time Management, Adaptability

LANGUAGES

- **Arabic:** Native

- **English:** Fluent
- **French:** Professional working proficiency

INTERESTS

Video Games | Hackathons & Challenges | Music